

AI & GENERATIVE AI PROGRAM

The AI / GenAI Lifecycle — With and Without Governance

A visual walkthrough of how a Generative AI system moves from idea to retirement — and how Governance, Responsible AI, and Security change the outcome at every stage.

GenAI Lifecycle Stages

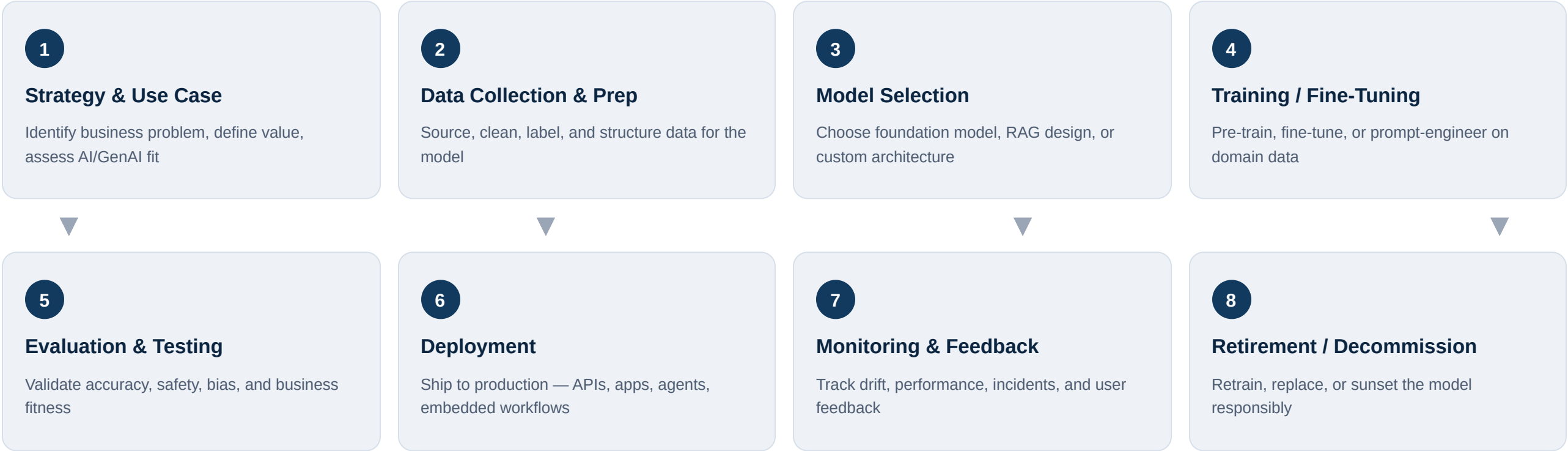
Without Governance — Risk View

With Governance — Controlled View

Security Overlay (MLSecOps)

The End-to-End GenAI Lifecycle

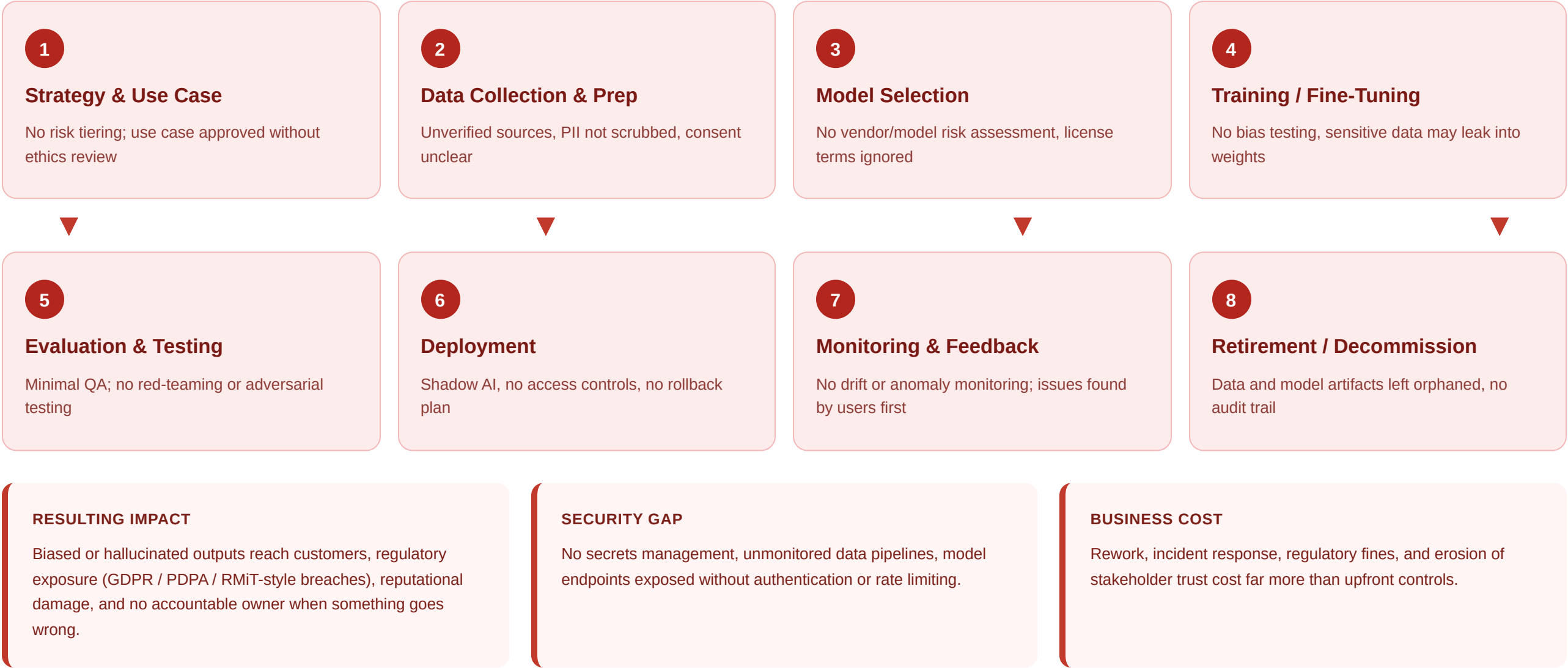
Eight stages every Generative AI initiative passes through — from defining the business use case to eventual retirement. This is the neutral, baseline flow before we layer in governance, responsible AI, and security controls.



Key idea: the stages themselves don't change whether or not an organization has governance. What changes is **what happens inside each stage** — the checks, owners, and controls applied. The next two pages show the same eight stages twice: once with governance absent, once with it built in.

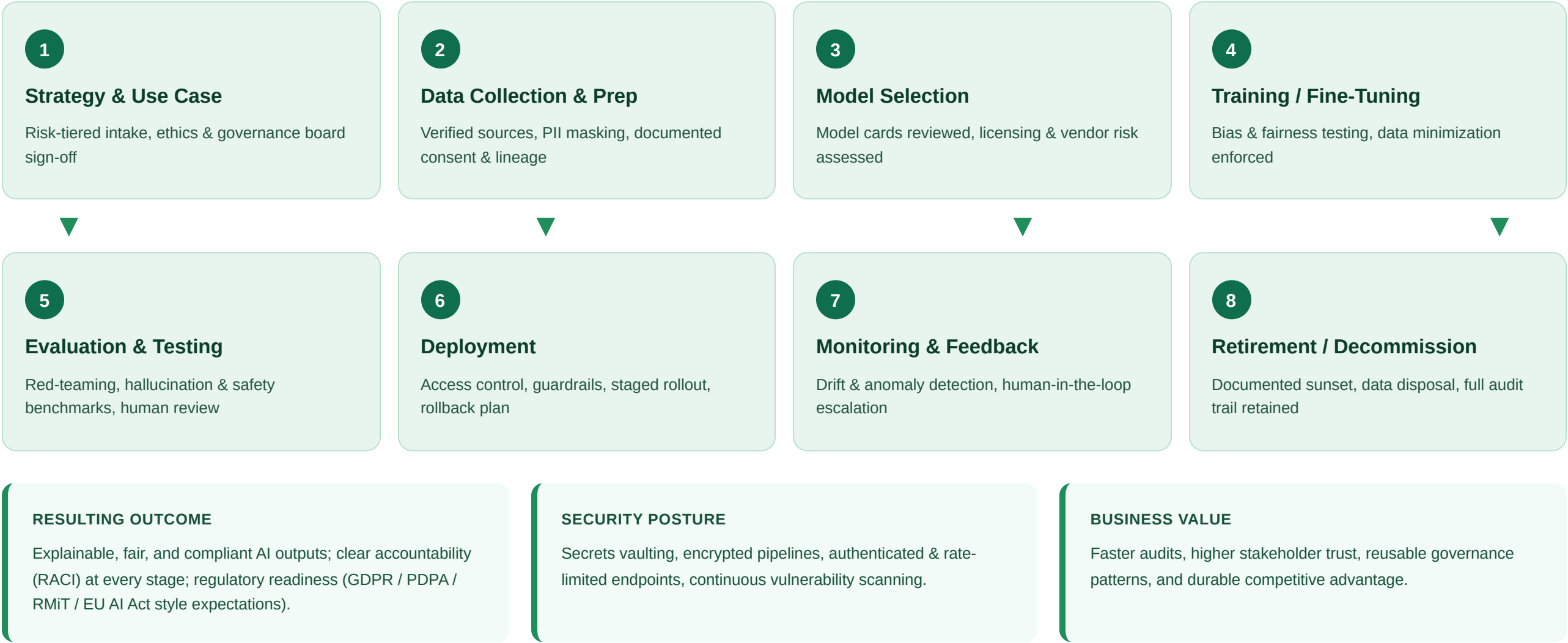
Lifecycle Without Governance — Where Risk Enters

Same eight stages, no oversight layer. Each stage silently accumulates risk that surfaces later as bias, hallucination, data leakage, compliance failure, or loss of trust.



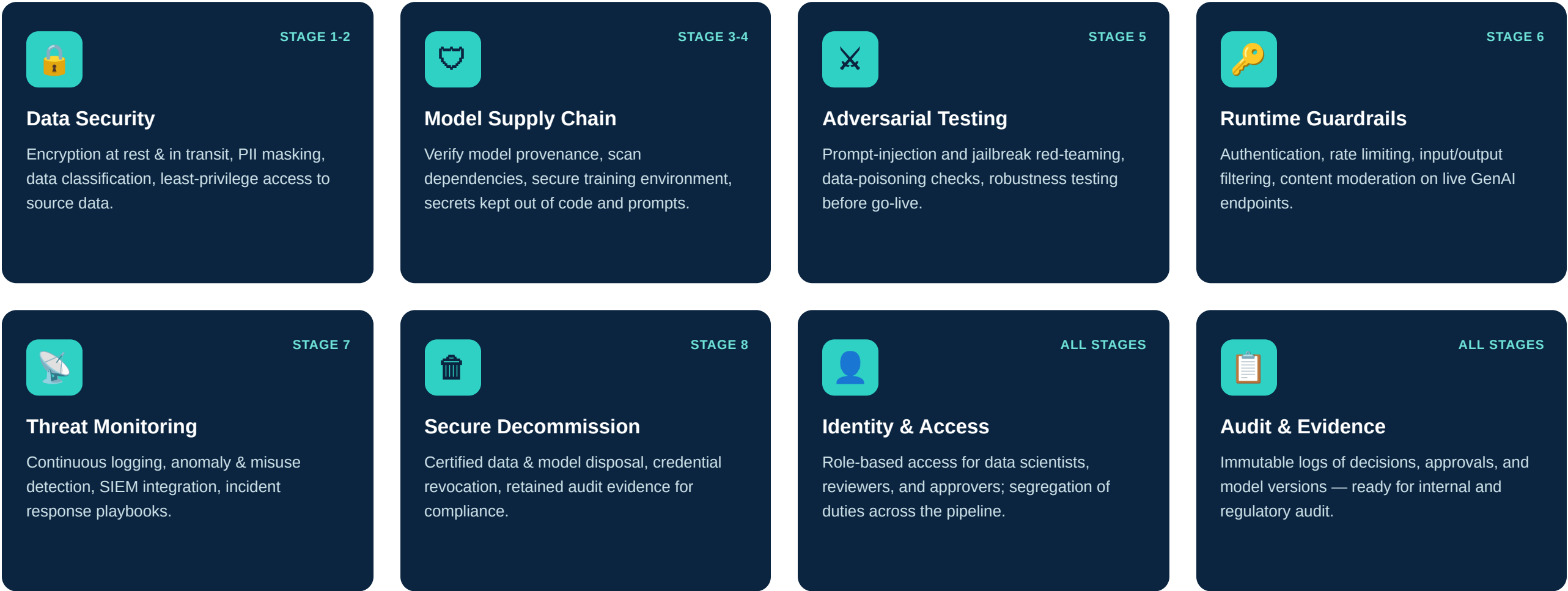
Lifecycle With Governance — Controls at Every Stage

The same eight stages, now with a governance and responsible-AI layer wrapped around each one — turning risk points into checkpoints.



Security Overlay Across the GenAI Lifecycle (MLSecOps)

Security isn't a single stage — it runs underneath all eight, from data ingestion to retirement. These are the controls that apply continuously.



Without Governance vs. With Governance & Responsible AI

The same lifecycle, compared across the dimensions that matter most to the business, regulators, and end users.

Dimension	Without Governance	With Governance & Responsible AI
Accountability	No clear owner; issues surface after harm is done	RACI defined per stage; named approvers and reviewers
Bias & Fairness	Untested; bias discovered by affected users or press	Fairness testing built into evaluation before deployment
Data Privacy	PII handling ad hoc; consent and lineage unclear	Documented consent, data lineage, and minimization
Security	Endpoints exposed, no red-teaming, weak secrets hygiene	Guardrails, adversarial testing, and access control by design
Transparency	"Black box" outputs, no explainability for decisions	Model cards, decision logs, explainable outputs
Regulatory Readiness	Reactive; scrambling when auditors or regulators ask	Audit trail and evidence maintained continuously
Cost Profile	Lower upfront cost, high downstream remediation cost	Modest upfront investment, lower lifetime risk cost
Stakeholder Trust	Erodes after first visible failure	Builds steadily through consistent, explainable outcomes

The Responsible AI Pillars That Hold Governance Together

These six pillars are what "governance" actually consists of in practice — applied consistently across every stage of the lifecycle.

● **Fairness**

Actively testing for and mitigating bias across protected groups and edge cases.

● **Transparency**

Explainable model behavior, documented via model cards and decision logs.

● **Accountability**

Named owners and approvers at every stage — no orphaned decisions.

● **Privacy**

Data minimization, consent management, and PII protection by design.

● **Safety**

Red-teaming, guardrails, and human-in-the-loop oversight for high-risk actions.

● **Security**

Protecting data, models, and endpoints from misuse across the full lifecycle.

Takeaway: governance and responsible AI aren't a separate phase bolted onto the end of a project — they are a layer that wraps every one of the eight lifecycle stages, from the first use-case conversation to the day the model is retired. Security runs underneath the whole thing. Organizations that build this in from stage 1 move faster later, because trust and compliance are already earned rather than retrofitted.